Problem 1

From lecture 5, slide 15

The log-likelihood for logistic regression is,

$$l_n(\beta) = \log(L_n(\beta)) = \sum_{i=1}^{n} y_i\left[x_i'\beta - \log(1 + e^{x_i'\beta})\right]$$

If we take partial derivative w.r.t β,

$$\frac{\partial l_n(\beta)}{\partial \beta} = \sum_{i=1}^{n} x_i' \left(y_i - \frac{e^{x_i'\beta}}{1 + e^{x_i'\beta}} \right)$$

We were given, a logistic regression with single binary covariate, x.

Here, $x_i'\beta = \beta_0 + \beta_1 x_i$ (one intercept and one x_i coefficient)

So,

$$\begin{pmatrix} \dfrac{\partial l_n(\beta)}{\partial \beta_0} \\ \\ \dfrac{\partial l_n(\beta)}{\partial \beta_1} \end{pmatrix} = \begin{pmatrix} \displaystyle\sum_{i=1}^{n} \left(y_i - \frac{e^{\beta_0 + \beta_1 x_i}}{1 + e^{\beta_0 + \beta_1 x_i}} \right) \\ \\ \displaystyle\sum_{i=1}^{n} x_i \left(y_i - \frac{e^{\beta_0 + \beta_1 x_i}}{1 + e^{\beta_0 + \beta_1 x_i}} \right) \end{pmatrix}$$

We need to simplify above partial derivatives to only involve $\bar{y}_0$ and $\bar{y}_1$.

$\bar{y}_0 \longrightarrow$ Average observed outcome among units with $x=0$

$\bar{y}_1 \longrightarrow$ Average observed outcome among units with $x=1$

$$\bar{y}_0 = \frac{1}{n_0} \sum_{i(x_i=0)} y_i \quad ; \quad \bar{y}_1 = \frac{1}{n_1} \sum_{i(x=1)} y_i$$

$$(n_0 + n_1 = n)$$

given, $$\sum_{i=1}^{n} c_i = \sum_{i=1}^{n} x_i c_i + \sum_{i=1}^{n} (1-x_i) c_i$$

$$= \sum_{i(x=0)} c_i + \sum_{i(x=1)} c_i$$

for $$\boxed{\sum_{i=1} y_i = n_0 \bar{y}_0 + n_1 \bar{y}_1}$$

multiplying with x_i, $$\sum_{i=1}^{n} x_i y_i = \sum_{i(x_i=1)} x_i y_i + \sum_{i(x_i=0)} x_i y_i$$

0

$$= \sum_{i(x_i=1)} y_i = n_1 \bar{y}_1$$

$$\begin{pmatrix} \frac{\partial \ln(\beta)}{\partial \beta_0} \\ \frac{\partial \ln(\beta)}{\partial \beta_1} \end{pmatrix} = \begin{pmatrix} \sum_{i=1}^{n} y_i \checkmark - \sum_i \frac{e^{\beta_0 + \beta_1 x_i}}{1 + e^{\beta_0 + \beta_1 x_i}} \\ \sum_i x_i y_i \checkmark - \sum x_i \frac{e^{\beta_0 + \beta_1 x_i}}{1 + e^{\beta_0 + \beta_1 x_i}} \end{pmatrix}$$

for $\sum_i \frac{e^{\beta_0+\beta_1 x_i}}{1+e^{\beta_0+\beta_1 x}} = \sum_{i(x_i=0)} \frac{e^{\beta_0}}{1+e^{\beta_0}} + \sum_{i(x_i=1)} \frac{e^{\beta_0+\beta_1}}{1+e^{\beta_0+\beta_1}}$

$$= n_0 \frac{e^{\beta_0}}{1+e^{\beta_0}} + n_1 \frac{e^{\beta_0+\beta_1}}{1+e^{\beta_0+\beta_1}}$$

$$\sum_i x_i \frac{e^{\beta_0+\beta_1 x_i}}{1+e^{\beta_0+\beta_1 x_i}} = \sum_{i(x_i=1)} \frac{e^{\beta_0+\beta_1 \cancel{x_i}}}{1+e^{\beta_0+\beta_1}} + 0$$

$$= n_1 \frac{e^{\beta_0+\beta_1}}{1+e^{\beta_0+\beta_1}}$$

using all the above results,

$$\begin{pmatrix} \frac{\partial l_n(\beta)}{\partial \beta_0} \\ \frac{\partial l_n(\beta)}{\partial \beta_1} \end{pmatrix} = \begin{pmatrix} n_0 \bar{y}_1 + n_1 \bar{y}_1 - \left(\frac{n_0 e^{\beta_0}}{1+e^{\beta_0}} + n_1 \frac{e^{\beta_0+\beta_1}}{1+e^{\beta_0+\beta_1}} \right) \\ n_1 \bar{y}_1 - n_1 \frac{e^{\beta_0+\beta_1}}{1+e^{\beta_0+\beta_1}} \end{pmatrix}$$

part 3.c,

setting both partial derivatives to zero, we have

$$n_0\bar{y}_0 + n_1\bar{y}_1 - \left(\frac{n_0 e^{\beta_0}}{1+e^{\beta_0}} + n_1\frac{e^{\beta_0+\beta_1}}{1+e^{\beta_0+\beta_1}}\right) = 0$$

$$n_1\bar{y}_1 - n_1\frac{e^{\beta_0+\beta_1}}{1+e^{\beta_0+\beta_1}} = 0$$

$\longrightarrow$ subtracting these two, we get

$$\cancel{n_0}\bar{y}_0 - \cancel{n_0}\frac{e^{\beta_0}}{1+e^{\beta_0}} = 0$$

$$\bar{y}_0 = \frac{e^{\beta_0}}{1+e^{\beta_0}} = \text{expit}(\beta_0)$$

So, $\beta_0 = \text{logit}(\bar{y}_0)$

$$= \log\left(\frac{\bar{y}_0}{1-\bar{y}_0}\right) \longrightarrow (1)$$

$$\bar{y}_1 = \frac{e^{\beta_0+\beta_1}}{1+e^{\beta_0+\beta_1}} = \text{expit}(\beta_0+\beta_1)$$

$$\therefore \beta_0+\beta_1 = \text{logit}(\bar{y}_1) = \log\left(\frac{\bar{y}_1}{1-\bar{y}_1}\right)$$

$$\beta_1 = \log\left(\frac{\bar{y}_1}{1-\bar{y}_1}\right) - \log\left(\frac{\bar{y}_0}{1-\bar{y}_0}\right)$$

$$\beta_1 = \log\left(\frac{\bar{y}_1(1-\bar{y}_0)}{\bar{y}_0(1-\bar{y}_1)}\right)$$

part d,

$Y=1 \rightarrow$ student smokes

$Y=0 \rightarrow$ otherwise

$X=1 \rightarrow$ atleast one parent/guardian smokes

$X=0 \rightarrow$ otherwise

	students smoke ($Y=1$)	$Y=0$
$X=1$	90	238
$X=0$	21	651

$$\bar{y}_1 = \frac{90}{90+238} = \frac{90}{328} \approx 0.274$$

$$\bar{y}_0 = \frac{21}{21+651} = \frac{21}{672} \approx 0.031$$

$$\beta_0 = \log\left(\frac{0.031}{1-0.031}\right) = -3.44$$

$$\beta_1 = \cancel{\beta}\log\left(\frac{0.274}{1-0.274}\right) - \beta_0 = \begin{array}{r} -0.974 \\ +3.44 \\ = 2.462 \end{array}$$

Stat_HW4

2025-03-01

#Problem 1, Part D

```r
data <- data.frame(
  x = c(rep(1, 328), rep(0, 672)),
  y = c(rep(1, 90), rep(0, 238), rep(1, 21), rep(0, 651))
)
model <- glm(y ~ x, data = data, family = binomial)
summary(model)
```

```
##
## Call:
## glm(formula = y ~ x, family = binomial, data = data)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -3.4340     0.2217 -15.489   <2e-16 ***
## x             2.4615     0.2539   9.695   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 697.20  on 999  degrees of freedom
## Residual deviance: 572.35  on 998  degrees of freedom
## AIC: 576.35
##
## Number of Fisher Scoring iterations: 6
```

The values match with the calculated ones.

#Problem 2

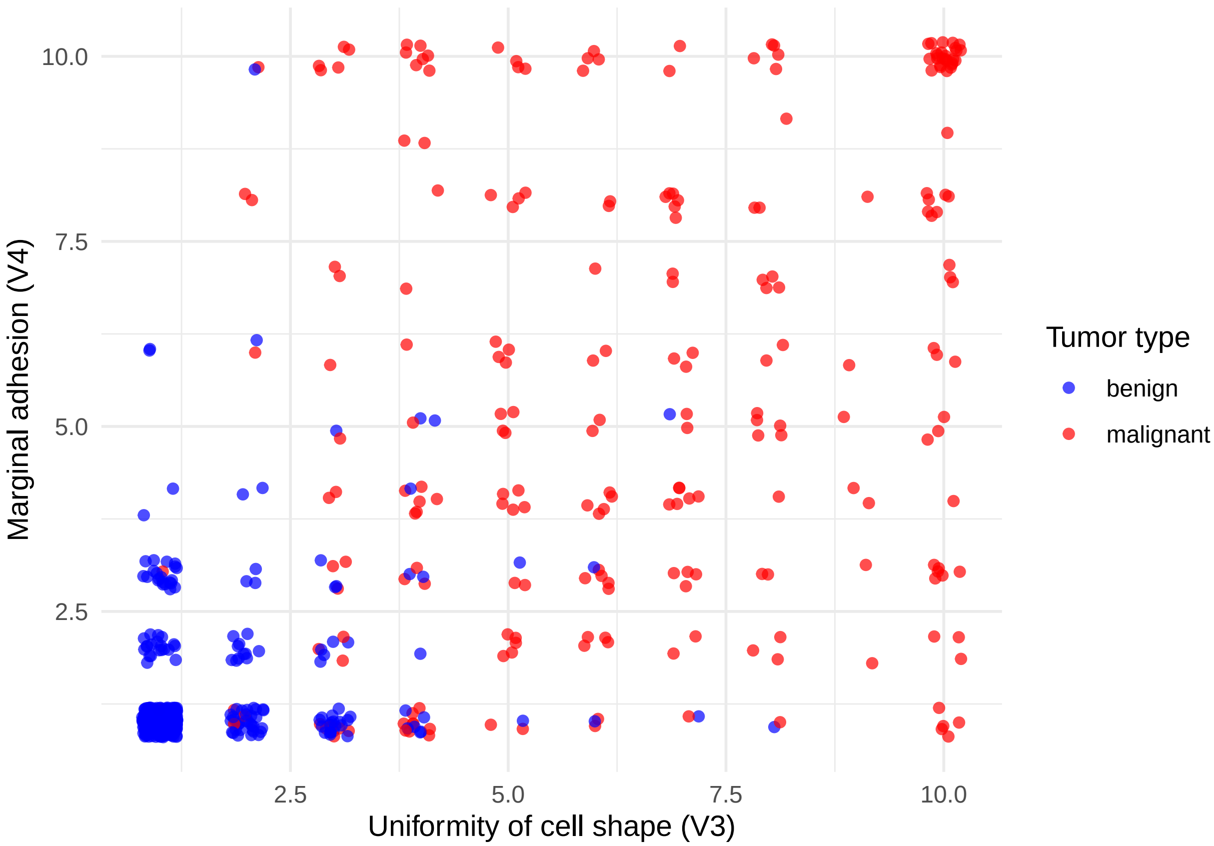
Part A

```r
library(MASS)
data(biopsy)
head(biopsy)
```

```
##        ID V1 V2 V3 V4 V5 V6 V7 V8 V9     class
## 1 1000025  5  1  1  1  2  1  3  1  1    benign
## 2 1002945  5  4  4  5  7 10  3  2  1    benign
## 3 1015425  3  1  1  1  2  2  3  1  1    benign
## 4 1016277  6  8  8  1  3  4  3  7  1    benign
## 5 1017023  4  1  1  3  2  1  3  1  1    benign
## 6 1017122  8 10 10  8  7 10  9  7  1 malignant
```

```
#?biopsy
```

```
biopsy_sp <- biopsy[complete.cases(biopsy),]
library(ggplot2)
ggplot(biopsy_sp, aes(x = V3, y = V4, color = class)) +
  geom_point(position = position_jitter(width = 0.2, height = 0.2), alpha = 0.7) +
  labs(x = "Uniformity of cell shape (V3)", y = "Marginal adhesion (V4)", color = "Tumor type") +
  scale_color_manual(values = c("blue", "red")) +
  theme_minimal()
```



As seen above, Benign tumors tend to have lower values of V3 and V4, they form a cluster. malignant tumors are more spread out. if someone has higher values of V3 and V4 we can say they have a malignant tumor.

Part B

```
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following object is masked from 'package:MASS':
##
##     select

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
```

```
##
##     intersect, setdiff, setequal, union
```

```
biopsy_b <- mutate(biopsy_sp, class2 = as.numeric(class) - 1) #benign (0), Malignant (1)
model <- glm(class2 ~ . -ID -class, data = biopsy_b, family = binomial)
summary(model)
```

```
##
## Call:
## glm(formula = class2 ~ . - ID - class, family = binomial, data = biopsy_b)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -10.10394    1.17488  -8.600  < 2e-16 ***
## V1            0.53501    0.14202   3.767 0.000165 ***
## V2           -0.00628    0.20908  -0.030 0.976039
## V3            0.32271    0.23060   1.399 0.161688
## V4            0.33064    0.12345   2.678 0.007400 **
## V5            0.09663    0.15659   0.617 0.537159
## V6            0.38303    0.09384   4.082 4.47e-05 ***
## V7            0.44719    0.17138   2.609 0.009073 **
## V8            0.21303    0.11287   1.887 0.059115 .
## V9            0.53484    0.32877   1.627 0.103788
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 884.35  on 682  degrees of freedom
## Residual deviance: 102.89  on 673  degrees of freedom
## AIC: 122.89
##
## Number of Fisher Scoring iterations: 8
```

Part C

```
beta_hat <- coef(model)
x_betahat <- as.matrix(biopsy_b[, -c(1, which(colnames(biopsy_b) %in% c("class", "class2")))]) %*% beta_
exp_function <- exp(x_betahat) / (1 + exp(x_betahat))
print(paste("Sum:",sum(biopsy_b$V1 * (biopsy_b$class2 - exp_function))))
```

```
## [1] "Sum: 9.94663054698641e-09"
```

This is approximately zero.

Part D

```
#V4 is margial adhesion
print(paste("Estimated coefficient for marginal adhesion:",coef(model)["V4"]))
```

```
## [1] "Estimated coefficient for marginal adhesion: 0.330636915299498"
```

It is positive so, increase in marginal adhesion will lead to more chances of having malignant tumor. 1 Unit increase in coefficient of V4, leads to $(1 - e^(0.3306))$ that is 39.2% increase in chances of havign a magglignant tumor.

Part E

```r
biopsy_e <- mutate(biopsy_sp, class2 = as.numeric(class) - 1)
biopsy_e <- biopsy_e[, !(colnames(biopsy_e) %in% c("ID", "class"))]
model <- glm(class2 ~ ., data = biopsy_e, family = binomial)
mean_values <- colMeans(biopsy_e[, names(coef(model))[-1], drop = FALSE])
mean_values_df <- as.data.frame(t(mean_values))
colnames(mean_values_df) <- names(coef(model))[-1]
pre_prob <- predict(model, newdata = mean_values_df, type = "response")
print(paste( "probability that the tumor is malignant",pre_prob))
```

```
## [1] "probability that the tumor is malignant 0.250838631639123"
```

Part F

Test Statistic,

$$Z = \frac{\hat{\beta}_j^2}{\mathrm{SE}(\hat{\beta}_j)^2}$$

For $H_0 : \beta_2 = \beta_3 = 0$,

$$Z_{\text{total}} = \frac{\hat{\beta}_2^2}{\mathrm{SE}(\hat{\beta}_2)^2} + \frac{\hat{\beta}_3^2}{\mathrm{SE}(\hat{\beta}_3)^2}$$

For P-Value,

$$p = 1 - P\left(\chi_2^2 \leq Z_{\text{total}}\right)$$

```r
Z2 <- (-0.00628)^2 / (0.20908)^2
Z3 <- (0.32271)^2 / (0.23060)^2
Z_total <- Z2 + Z3
p_value <- 1 - pchisq(Z_total, df = 2)
print(p_value)
```

```
## [1] 0.375438
```

P-value is greater than 0.05, so we fail to reject null hypothesis. V2 and V3 are not significant.

Part G Just like earlier,

$$Z_9 = \frac{\hat{\beta}_9^2}{\mathrm{SE}(\hat{\beta}_9)^2} = \frac{(0.53484)^2}{(0.32877)^2} = 2.646$$

$$p = 1 - P\left(\chi_1^2 \leq Z_9\right) = 0.1038$$

The p-value is slightly greater than 0.1, so we fail to reject null hypothesis. mitosis doesn't significatnly effect odds of tumor being malginant.

Part H

$$CI = \hat{\beta}_1 \pm Z_{\text{critical}} \cdot \mathrm{SE}(\hat{\beta}_1) = 0.53501 \pm 1.96 \times 0.14202 = (0.2567, 0.8134)$$

This does't include zero, so V1 is stastically significant. If we repeat the experiemnt, 95% of the time the coefficeint of V1 will be in this interval. as both bounds are positive, higher V1 means more odds of having a malignant tumor.

#Problem 3

Part A

I will be using sonar dataset in mlbench package available in R. The Sonar dataset is a data frame with 208 observations on 61 variables. The label with other observation has R for Rock or M for Mine (binary variable for outcome, mine =1, rock =0). Each number (quantitative varialbes, we have 60) represents the energy within a particular frequency band, integrated over a certain period of time basically sonar frequency reflection values.

```
#install.packages("mlbench")
library(mlbench)
data(Sonar)
sonar_df <- Sonar
head(sonar_df)
```

```
##       V1     V2     V3     V4     V5     V6     V7     V8     V9    V10    V11
## 1 0.0200 0.0371 0.0428 0.0207 0.0954 0.0986 0.1539 0.1601 0.3109 0.2111 0.1609
## 2 0.0453 0.0523 0.0843 0.0689 0.1183 0.2583 0.2156 0.3481 0.3337 0.2872 0.4918
## 3 0.0262 0.0582 0.1099 0.1083 0.0974 0.2280 0.2431 0.3771 0.5598 0.6194 0.6333
## 4 0.0100 0.0171 0.0623 0.0205 0.0205 0.0368 0.1098 0.1276 0.0598 0.1264 0.0881
## 5 0.0762 0.0666 0.0481 0.0394 0.0590 0.0649 0.1209 0.2467 0.3564 0.4459 0.4152
## 6 0.0286 0.0453 0.0277 0.0174 0.0384 0.0990 0.1201 0.1833 0.2105 0.3039 0.2988
##      V12    V13    V14    V15    V16    V17    V18    V19    V20    V21    V22
## 1 0.1582 0.2238 0.0645 0.0660 0.2273 0.3100 0.2999 0.5078 0.4797 0.5783 0.5071
## 2 0.6552 0.6919 0.7797 0.7464 0.9444 1.0000 0.8874 0.8024 0.7818 0.5212 0.4052
## 3 0.7060 0.5544 0.5320 0.6479 0.6931 0.6759 0.7551 0.8929 0.8619 0.7974 0.6737
## 4 0.1992 0.0184 0.2261 0.1729 0.2131 0.0693 0.2281 0.4060 0.3973 0.2741 0.3690
## 5 0.3952 0.4256 0.4135 0.4528 0.5326 0.7306 0.6193 0.2032 0.4636 0.4148 0.4292
## 6 0.4250 0.6343 0.8198 1.0000 0.9988 0.9508 0.9025 0.7234 0.5122 0.2074 0.3985
##      V23    V24    V25    V26    V27    V28    V29    V30    V31    V32    V33
## 1 0.4328 0.5550 0.6711 0.6415 0.7104 0.8080 0.6791 0.3857 0.1307 0.2604 0.5121
## 2 0.3957 0.3914 0.3250 0.3200 0.3271 0.2767 0.4423 0.2028 0.3788 0.2947 0.1984
## 3 0.4293 0.3648 0.5331 0.2413 0.5070 0.8533 0.6036 0.8514 0.8512 0.5045 0.1862
## 4 0.5556 0.4846 0.3140 0.5334 0.5256 0.2520 0.2090 0.3559 0.6260 0.7340 0.6120
## 5 0.5730 0.5399 0.3161 0.2285 0.6995 1.0000 0.7262 0.4724 0.5103 0.5459 0.2881
## 6 0.5890 0.2872 0.2043 0.5782 0.5389 0.3750 0.3411 0.5067 0.5580 0.4778 0.3299
##      V34    V35    V36    V37    V38    V39    V40    V41    V42    V43    V44
## 1 0.7547 0.8537 0.8507 0.6692 0.6097 0.4943 0.2744 0.0510 0.2834 0.2825 0.4256
## 2 0.2341 0.1306 0.4182 0.3835 0.1057 0.1840 0.1970 0.1674 0.0583 0.1401 0.1628
## 3 0.2709 0.4232 0.3043 0.6116 0.6756 0.5375 0.4719 0.4647 0.2587 0.2129 0.2222
## 4 0.3497 0.3953 0.3012 0.5408 0.8814 0.9857 0.9167 0.6121 0.5006 0.3210 0.3202
## 5 0.0981 0.1951 0.4181 0.4604 0.3217 0.2828 0.2430 0.1979 0.2444 0.1847 0.0841
## 6 0.2198 0.1407 0.2856 0.3807 0.4158 0.4054 0.3296 0.2707 0.2650 0.0723 0.1238
##      V45    V46    V47    V48    V49    V50    V51    V52    V53    V54    V55
## 1 0.2641 0.1386 0.1051 0.1343 0.0383 0.0324 0.0232 0.0027 0.0065 0.0159 0.0072
## 2 0.0621 0.0203 0.0530 0.0742 0.0409 0.0061 0.0125 0.0084 0.0089 0.0048 0.0094
## 3 0.2111 0.0176 0.1348 0.0744 0.0130 0.0106 0.0033 0.0232 0.0166 0.0095 0.0180
## 4 0.4295 0.3654 0.2655 0.1576 0.0681 0.0294 0.0241 0.0121 0.0036 0.0150 0.0085
## 5 0.0692 0.0528 0.0357 0.0085 0.0230 0.0046 0.0156 0.0031 0.0054 0.0105 0.0110
## 6 0.1192 0.1089 0.0623 0.0494 0.0264 0.0081 0.0104 0.0045 0.0014 0.0038 0.0013
##      V56    V57    V58    V59    V60 Class
## 1 0.0167 0.0180 0.0084 0.0090 0.0032     R
## 2 0.0191 0.0140 0.0049 0.0052 0.0044     R
```

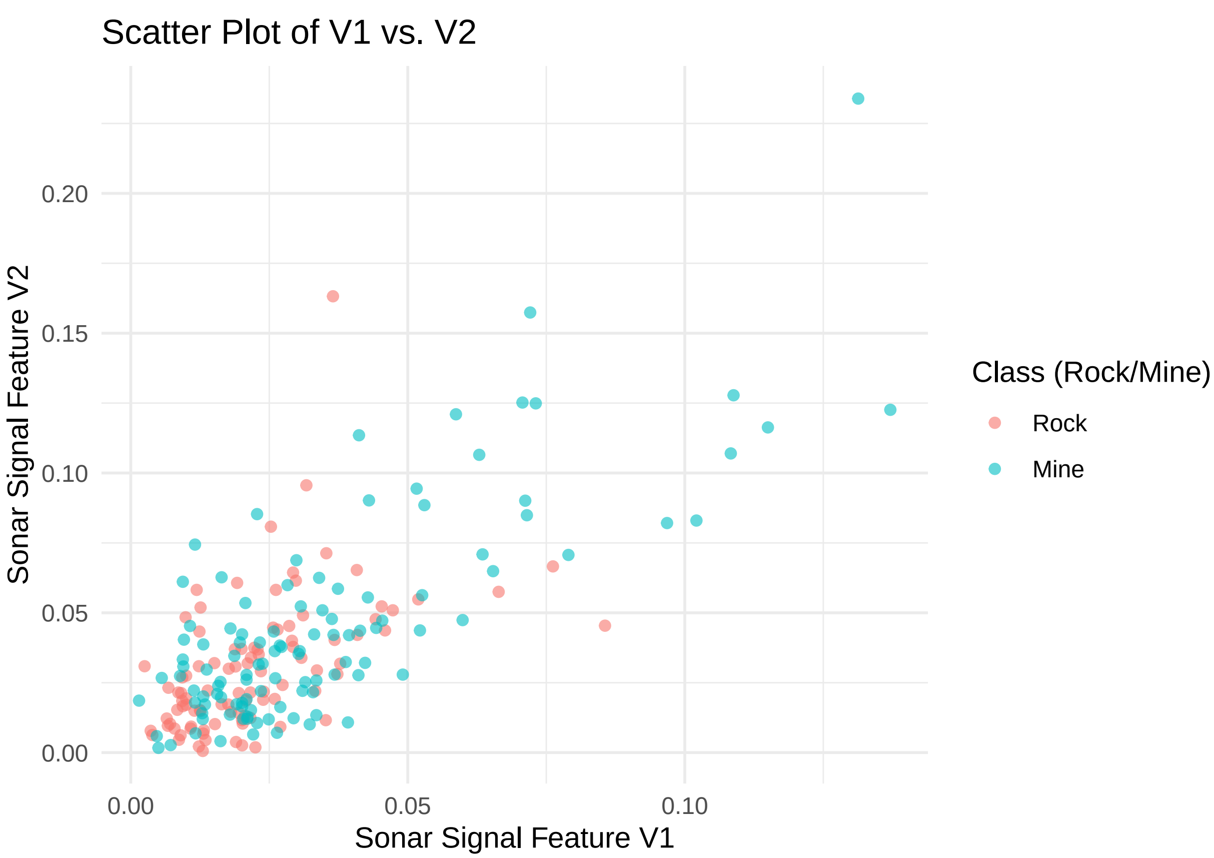
```
## 3 0.0244 0.0316 0.0164 0.0095 0.0078     R
## 4 0.0073 0.0050 0.0044 0.0040 0.0117     R
## 5 0.0015 0.0072 0.0048 0.0107 0.0094     R
## 6 0.0089 0.0057 0.0027 0.0051 0.0062     R
```

```r
sonar_df$Class <- as.numeric(sonar_df$Class == "M")
sonar_df <- na.omit(sonar_df)
```

Part B

I chose V1 and V2.

```r
library(ggplot2)
sonar_df$Class <- factor(sonar_df$Class, levels = c(0, 1), labels = c("Rock", "Mine"))
ggplot(sonar_df, aes(x = V1, y = V2, color = Class)) +
  geom_point(alpha = 0.6) +
  labs(
    title = "Scatter Plot of V1 vs. V2",
    x = "Sonar Signal Feature V1",
    y = "Sonar Signal Feature V2",
    color = "Class (Rock/Mine)"
  ) +
  theme_minimal()
```



Part C

```r
model <- glm(Class ~ V1 + V2 + V3, data = sonar_df, family = binomial)
summary(model)
```

```
##
## Call:
## glm(formula = Class ~ V1 + V2 + V3, family = binomial, data = sonar_df)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -0.7966     0.2808  -2.836  0.00456 **
## V1           25.5102    11.0831   2.302  0.02135 *
## V2            3.3775     8.8171   0.383  0.70167
## V3            2.5853     6.6095   0.391  0.69569
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 287.41  on 207  degrees of freedom
## Residual deviance: 268.94  on 204  degrees of freedom
## AIC: 276.94
##
## Number of Fisher Scoring iterations: 4
```

We fit model on V1, V2 and V3. V1 coeff is positve, meaning higher value of V1 means higher chances of detecting a mine. P-value is less than 0.05, meaning V1 is statistically significant.

Part D

```r
p_value_V2 <- summary(model)$coefficients["V2", "Pr(>|z|)"]
print(p_value_V2)
```

```
## [1] 0.7016745
```

P-value is grteater than 0.05, meaning V2 is not statistically significant. We fail to reject null hypothesis that V2 has no effect on the outcome.

Part E

```r
confint(model, "V3", level = 0.95)
```

```
## Waiting for profiling to be done...

##     2.5 %    97.5 %
## -10.25748  15.80286
```

Lower bound is -10.2578 means that sometimes, if V3 value increases the odds of finding a mine will decrease. Upper bound is 15.8026 means that sometimes, if V3 value increases the odds of finding a mine will increase. As, this does include zero, V3 is not statistically significant, it doesn't have any effect on outcome.